

Sri Lanka Institute of Information Technology

IT3071 - Machine Learning and Optimization Methods

Assignment 02 - Research Report

Next Intelligence: Designing the Next Generation of Artificial Intelligence Inspired by the Human Mind

Selected Capability: **Unconscious Error Detection**
Proposed Framework: Preconscious Error Intelligence Framework (PEIF)

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1. Abstract

Artificial Intelligence has progressed from traditional Machine Learning to Deep Learning, Large Language Models (LLMs), and Agentic AI systems. Despite these advancements, existing AI systems often rely on explicit evaluation mechanisms, external feedback, or verification processes to identify and correct errors. In contrast, human cognition can sometimes recognize that something may be wrong before consciously identifying the exact mistake. This capability, known as **Unconscious Error Detection**, provides the inspiration for this research.

This study investigates the scientific foundations of unconscious error detection, evaluates related limitations of current AI approaches, and proposes the **Preconscious Error Intelligence Framework (PEIF)** as a conceptual architecture for future Artificial Intelligence. PEIF introduces continuous internal cognitive-state monitoring, expectation-based coherence evaluation, and a graded warning state called **Cognitive Tension**. This warning can trigger adaptive attention and diagnostic reasoning before the exact cause of a potential error is identified. The framework also proposes experience-based learning through Error Trace Memory.

Rather than developing another AI application, PEIF presents a brain-inspired architectural approach to early error sensing and adaptive cognitive monitoring. Its potential contributions include improved AI reliability, more efficient error investigation, and safer decision-making, subject to future implementation and experimental validation.

2. . Introduction

Artificial Intelligence (AI) has achieved remarkable progress in perception, prediction, natural language processing, reasoning, and autonomous decision-making. Technologies such as Deep Learning, Large Language Models (LLMs), and Agentic AI have enabled intelligent systems to perform increasingly complex tasks across various domains. However, despite these developments, current AI systems still represent only a limited range of the higher-order cognitive capabilities demonstrated by the human mind.

One such capability is **Unconscious Error Detection**, where an individual may recognize that something is wrong before consciously understanding the exact mistake. For example, a person solving a mathematical problem may experience a sense of uncertainty or inconsistency before identifying an incorrect calculation. Research in cognitive psychology and neuroscience suggests that human performance monitoring involves early error-related processing, attention regulation, and subsequent conscious evaluation.

Current AI systems already possess mechanisms such as anomaly detection, uncertainty estimation, output verification, and self-correction. However, integrating these mechanisms into a continuous internal cognitive-monitoring architecture that can generate a pre-diagnostic warning before identifying the exact error remains an important research challenge.

To explore this gap, this study proposes the **Preconscious Error Intelligence Framework (PEIF)**, a conceptual architecture inspired by human error-monitoring processes. The framework introduces internal cognitive-state monitoring, an expectation model, a graded warning state called **Cognitive Tension**, adaptive attention control, diagnostic reasoning, and experience-based learning.

The main objective of this research is to investigate how an AI system could recognize possible instability within its own cognitive processing and respond appropriately before the exact cause of an error is known. The study examines the scientific inspiration, limitations of existing AI, proposed computational principles, conceptual architecture, feasibility, and ethical implications of PEIF.

Rather than claiming to reproduce human consciousness, PEIF proposes a functional approach to pre-diagnostic error monitoring that could contribute to the development of more reliable, adaptive, and self-monitoring AI systems.

3. Part A – Understanding the Human Mind: Unconscious Error Detection

3.1. Definition and Characteristics

Unconscious Error Detection is the ability of the human mind to notice that something may be wrong **before the person fully understands the mistake consciously**. In simple terms, a person may feel that an answer, action, or decision is incorrect even though they cannot immediately explain why.

For example, a student may solve a calculation and look at the answer:

"This answer does not look correct."

At this stage, the student may not know which step is wrong or what the correct answer is. However, the brain has already detected some kind of conflict or mismatch.

The general process can be represented as:

Action or thought → Internal monitoring → Mismatch detected → Error signal → Attention → Conscious awareness → Correction.

The main characteristics of unconscious error detection are that it can happen very quickly, it can occur without external feedback, it involves monitoring our own behaviour, and it can help us change or correct future actions. It is also important to understand that detecting a possible error does not mean that the person already knows the correct answer.

3.2. Scientific Understanding

Unconscious error detection is closely related to a scientific concept called **performance monitoring**.

Performance monitoring is the process by which the brain continuously checks whether our actions are producing the expected result.

A simple model is:

Goal → Action → Monitor result → Detect problem → Adjust behaviour.

Scientists have proposed several explanations for how the brain may detect errors.

One explanation is the **mismatch or comparator idea**. The brain may compare what we intended to do with what actually happened.

For example:

Intended action: Press the right button.
Actual action: Press the left button.

Because the intended and actual actions are different, the brain may generate an error-related signal.

Another explanation is **conflict monitoring**. Sometimes two possible responses are active at the same time. When they conflict with each other, the brain may recognize that something is wrong.

The third explanation is related to **prediction error**. The brain expects one result, but a different result happens.

For example:

Expected result: Success
Actual result: Failure

The difference between the expected and actual result can become a signal that helps the person learn and change future behaviour.

3.3. Psychological Perspectives

From a psychological point of view, unconscious error detection is connected to **metacognition**.

Metacognition means:

Thinking about and monitoring our own thinking.

For example, normal thinking may be:

“I think this answer is correct.”

Metacognitive thinking may be:

“How sure am I about this answer?”
“Did I make a mistake?”

Humans are therefore able not only to perform a task, but also to evaluate how well they performed it.

Another related idea is **confidence and uncertainty**. A person may give an answer but feel uncertain about it. However, low confidence does not always mean the answer is wrong, and high confidence does not always mean the answer is correct.

Attention is also important. A possible error may first be processed in the background. If the problem becomes important enough, attention can move toward it.

For example:

Background monitoring → Possible problem → Attention → Deliberate checking

A student may suddenly think:

“That sentence does not sound right.”

Only after checking it carefully may the student find the actual grammar or logic error.

People can also change their behaviour after making errors. For example, after a mistake, they may slow down and become more careful. This is one example of how error detection can support learning and adaptation

3.4. Neuroscientific Evidence

Researchers study error detection using methods such as **EEG – Electroencephalography**.

EEG measures electrical activity from the brain using electrodes placed on the scalp. Researchers can compare brain activity during correct responses and incorrect responses.

Two important brain signals related to errors are:

3.4.1. Error-Related Negativity – ERN

The **ERN** is a very fast brain response that appears around the time a person makes an error. It can reach its maximum very shortly after the incorrect response.

This is important because it suggests that the brain can begin processing an error very quickly, sometimes before the person clearly reports being aware of it.

3.4.2. Error Positivity – Pe

The **Pe** happens later than the ERN and is more strongly connected with conscious awareness of the mistake.

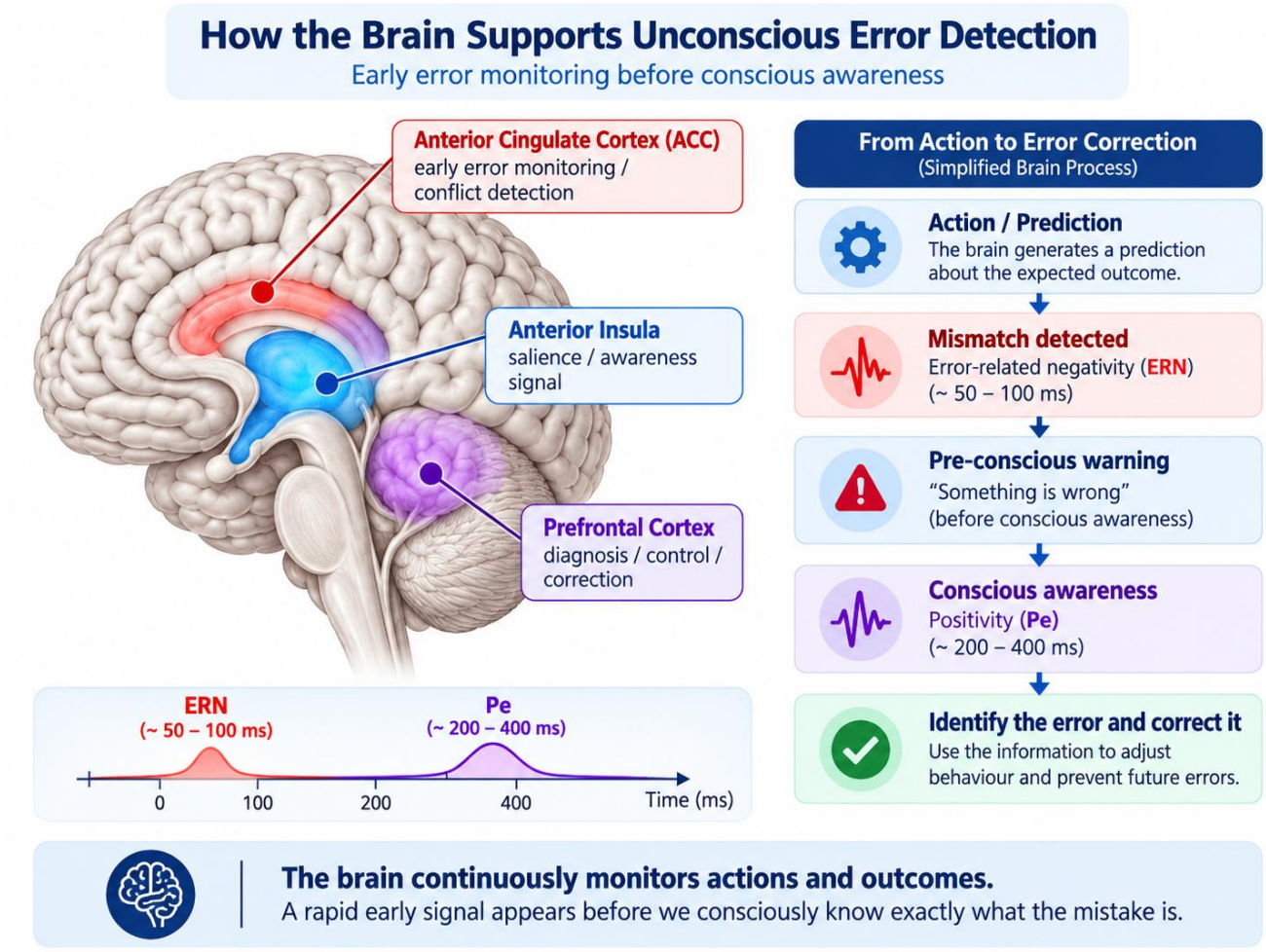
A simplified idea is:

Error → Early error processing → More evidence → Conscious awareness

However, it is important not to say:

ERN = unconscious
and
Pe = conscious

The real relationship is more complex and is still being researched.



3.4.3. Anterior Cingulate Cortex – ACC

The **Anterior Cingulate Cortex**, or ACC, and nearby medial frontal brain areas are often associated with conflict monitoring and performance monitoring.

A simple explanation is:

Conflict or error → Monitoring system reacts → More attention/control → Behaviour changes

However, error processing does not happen in only one area of the brain. It involves several connected brain regions.

3.5. Philosophical Interpretations

Unconscious error detection creates an interesting philosophical question:

Can a person detect something without consciously knowing that they detected it?

The important idea is that **processing information is not the same as consciously experiencing that information**.

The brain may process:

“My action does not match my intention.”

before the person consciously thinks:

“I made a mistake.”

Another important distinction is:

“Something is wrong”

versus

“I know exactly what is wrong.”

A person may detect that an answer is incorrect without knowing the exact reason.

Therefore, there may be several stages:

Detect inconsistency → Recognize error → Find the cause → Find the correct solution.

This idea is very important for your future AI framework because an AI system may not need to immediately know the correct answer. It may first produce an internal warning such as:

“My current decision may be unreliable and needs to be checked.”

3.6. Why It Is Considered Uniquely Human

Basic error detection is **not completely unique to humans**. Research has found error-monitoring behaviour and related brain activity in some non-human primates as well.

Therefore, your report should not say:

“Only humans can detect errors.”

The stronger point is that humans can combine error detection with many higher-level abilities such as:

language, abstract reasoning, memory, self-reflection, explanation, planning, conscious awareness, and deliberate correction.

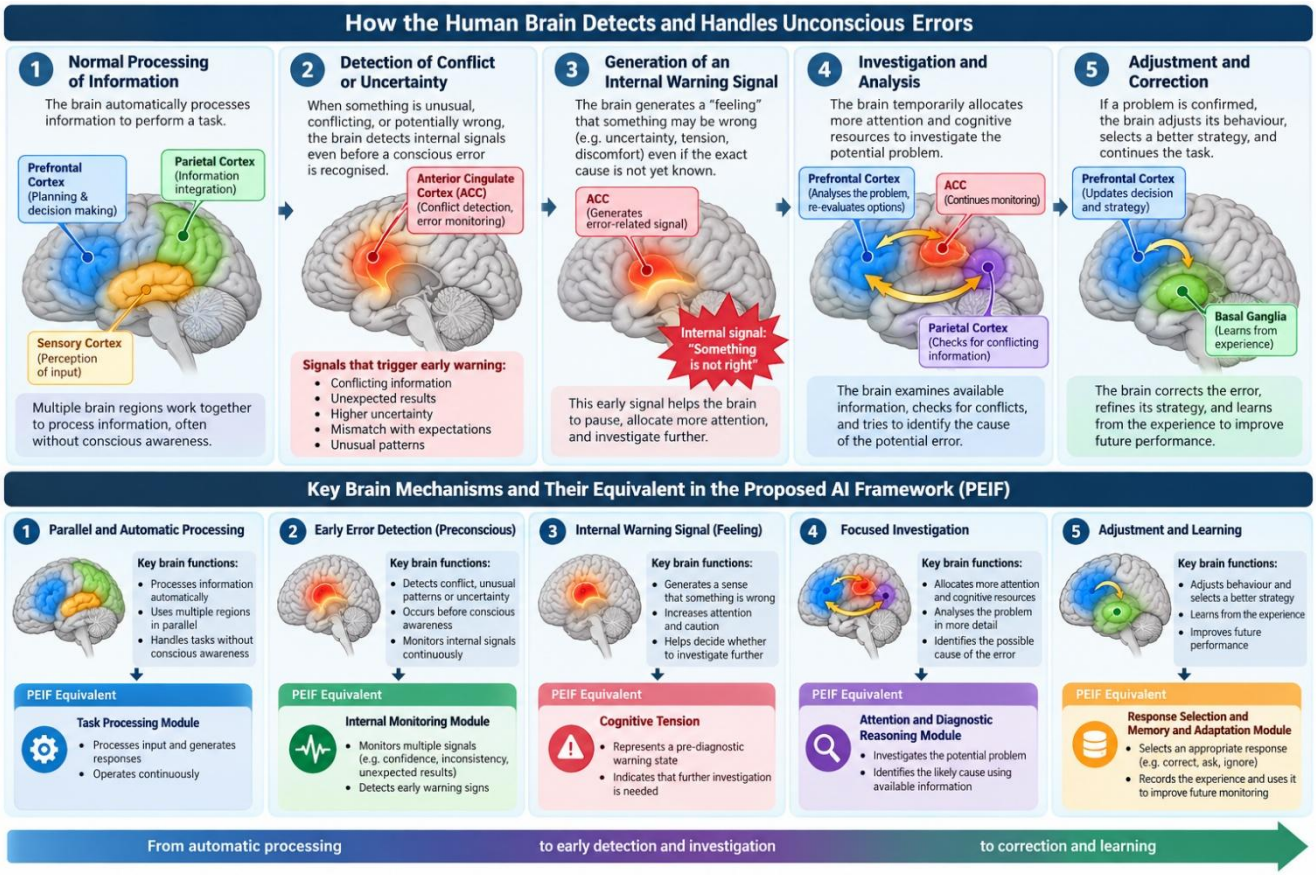
For example, a person can say:

“I think my answer is wrong because it conflicts with something I assumed earlier. I need to check the previous step.”

That involves more than detecting an error. It combines:

Error detection + awareness + memory + reasoning + language + self-reflection + correction.

Therefore, the important human capability for this project is not simple error detection, but **advanced internal error monitoring combined with conscious reflection and reasoning**.


How the Human Brain Detects and Handles Unconscious Errors
1 Normal Processing of Information
The brain automatically processes information to perform a task.
Prefrontal Cortex (Planning & decision making)
Parietal Cortex (Information integration)
Sensory Cortex (Perception of input)
Multiple brain regions work together to process information, often without conscious awareness.
2 Detection of Conflict or Uncertainty
When something is unusual, conflicting, or potentially wrong, the brain detects internal signals even before a conscious error is recognised.
Anterior Cingulate Cortex (ACC) (Conflict detection, error monitoring)
Signals that trigger early warning:
Conflicting information
Unexpected results
Higher uncertainty
Mismatch with expectations
Unusual patterns
3 Generation of an Internal Warning Signal
The brain generates a "feeling" that something may be wrong (e.g. uncertainty, tension, discomfort) even if the exact cause is not yet known.
ACC (Generates error-related signal)
Internal signal: "Something is not right"
This early signal helps the brain to pause, allocate more attention, and investigate further.
4 Investigation and Analysis
The brain temporarily allocates more attention and cognitive resources to investigate the potential problem.
Prefrontal Cortex (Analyses the problem, re-evaluates options)
ACC (Continues monitoring)
Parietal Cortex (Checks for conflicting information)
The brain examines available information, checks for conflicts, and tries to identify the cause of the potential error.
5 Adjustment and Correction
If a problem is confirmed, the brain adjusts its behaviour, selects a better strategy, and continues the task.
Prefrontal Cortex (Updates decision and strategy)
Basal Ganglia (Learns from experience)
The brain corrects the error, refines its strategy, and learns from the experience to improve future performance.
Key Brain Mechanisms and Their Equivalent in the Proposed AI Framework (PEIF)
1 Parallel and Automatic Processing
Key brain functions:
Processes information automatically
Uses multiple regions in parallel
Handles tasks without conscious awareness
PEIF Equivalent
Task Processing Module
Processes input and generates responses
Operates continuously
2 Early Error Detection (Preconscious)
Key brain functions:
Detects conflict, unusual patterns or uncertainty
Occurs before conscious awareness
Monitors internal signals continuously
PEIF Equivalent
Internal Monitoring Module
Monitors multiple signals (e.g. confidence, inconsistency, unexpected results)
Detects early warning signs
3 Internal Warning Signal (Feeling)
Key brain functions:
Generates a sense that something is wrong
Increases attention and caution
Helps decide whether to investigate further
PEIF Equivalent
Cognitive Tension
Represents a pre-diagnostic warning state
Indicates that further investigation is needed
4 Focused Investigation
Key brain functions:
Allocates more attention and cognitive resources
Analyses the problem in more detail
Identifies the possible cause of the error
PEIF Equivalent
Attention and Diagnostic Reasoning Module
Investigates the potential problem
Identifies the likely cause using available information
5 Adjustment and Learning
Key brain functions:
Adjusts behaviour and selects a better strategy
Learns from the experience
Improves future performance
PEIF Equivalent
Response Selection and Memory and Adaptation Module
Selects an appropriate response (e.g. correct, ask, ignore)
Records the experience and uses it to improve future monitoring
From automatic processing
to early detection and investigation
to correction and learning

3.7. Existing Knowledge Gaps

Even though error detection has been studied for many years, several questions are still unanswered.

Scientists are still discussing what the ERN actually represents. It may represent an error, conflict, prediction difference, or a more general signal that additional control is needed.

Another major question is:

How does an early unconscious error signal become conscious awareness?

Researchers do not yet fully understand how the brain moves from:

“Something may be wrong”

to:

“I know I made an error.”

It is also still unclear exactly what the brain compares when detecting a mistake. It may compare intended and actual actions, expected and actual outcomes, or several competing responses at the same time.

Another gap is that detecting an error does not always mean that the person will successfully correct it. Researchers are still studying how error signals lead to actual behavioural changes.

Different kinds of errors may also use different mechanisms. Detecting a wrong button press may not be the same as detecting an error in mathematics, language, planning, or moral reasoning.

One of the biggest unanswered questions is:

Why does some error-related brain activity become the conscious feeling that “something is wrong,” while other error-related activity remains outside awareness?

This final knowledge gap is especially important for your assignment because it gives you the foundation for creating a new AI framework inspired by unconscious error detection.

4. Part B – Limitations of Current Artificial Intelligence

4.1. Capabilities of Current AI Systems

Current Artificial Intelligence systems are capable of performing complex tasks such as **prediction, classification, pattern recognition, natural language generation, planning, and decision-making**. Deep Learning systems can learn complex patterns from large datasets and make predictions without requiring every rule to be manually programmed. Large Language Models (LLMs) extend these capabilities by generating and analysing language, answering questions, and performing various reasoning tasks.

Current AI also has several mechanisms related to **error detection and correction**. Machine Learning systems can use **prediction errors and loss functions** during training to improve their performance. AI systems can also perform **anomaly detection** to identify unusual patterns, while uncertainty and confidence estimation techniques can indicate when a prediction may be unreliable.

LLMs can sometimes identify inconsistencies, review generated responses, and correct mistakes when they are prompted to reconsider their answers. Similarly, **Agentic AI systems** can plan actions, interact with an environment, use tools and memory, evaluate outcomes, and modify their actions when a task fails.

Therefore, current AI already demonstrates several capabilities related to detecting and responding to errors. However, these abilities do not necessarily mean that current AI possesses **Unconscious Error Detection**. This distinction is examined in the next section.

4.2. Do Current AI Systems Possess Unconscious Error Detection?

Current AI systems demonstrate some behaviours that **resemble parts of error monitoring**, but they do not genuinely possess **Unconscious Error Detection** as defined in this study. AI can detect anomalies, estimate uncertainty, verify outputs, and sometimes correct its own mistakes. However, these processes usually depend on **explicit evaluation mechanisms, predefined objectives, external feedback, or additional verification steps**.

The main difference is the separation between **sensing a possible error** and **identifying the exact error**. In humans, as discussed in Part A, the brain may begin processing a mismatch before the person consciously understands what went wrong. A person may first experience a sense that *"something is wrong"* and only later identify the actual mistake.

Current AI generally lacks an equivalent **continuous internal monitoring process** that observes its own cognitive state and produces such a pre-diagnostic warning. For example, an LLM may generate an incorrect answer confidently and continue its response unless it is prompted to reconsider, receives external feedback, or another mechanism evaluates its output.

Therefore, current AI can **detect and correct certain errors**, but this is not equivalent to Unconscious Error Detection. The missing capability is the ability to internally recognize that its own cognitive process may be becoming unreliable **before knowing exactly what the error is**.

4.3. Why Existing AI Architectures Fail to Achieve Unconscious Error Detection

Existing AI architectures are mainly designed to **process inputs and produce accurate outputs**, rather than continuously monitor the condition of their own internal processing. In Deep Learning, errors are commonly identified through loss functions by comparing predicted and expected outputs. This helps the model learn during training, but it does not provide an internal mechanism that independently senses that its current cognitive process may be going wrong.

Similarly, LLMs can review or correct their responses, but this usually requires **additional prompting, verification, external feedback, or another evaluation process**. Agentic AI can monitor task outcomes and reflect on unsuccessful actions, but such reflection is generally triggered by observable results, predefined workflows, or feedback from the environment.

Another limitation is that existing architectures do not generally combine multiple internal signals—such as **prediction instability, confidence changes, memory conflicts, goal conflicts, and expectation mismatches**—into a separate pre-diagnostic state representing *"something may be wrong."*

Therefore, existing architectures mainly focus on **detecting identifiable errors or evaluating outcomes**, rather than creating an intermediate internal warning before the exact error is known. This architectural gap prevents current AI from fully achieving the Unconscious Error Detection capability explored in this study.

4.4. Fundamental Assumptions and Limitations of Current AI Paradigms

Most current AI paradigms are built around the assumption that intelligence can be achieved by **learning patterns, optimizing objectives, and producing accurate outputs**. Deep Learning models primarily optimize loss functions, while LLMs generate responses from learned statistical patterns. Agentic AI extends these systems with planning, memory, tool use, and feedback-driven actions. However, these approaches still mainly focus on **task performance rather than continuous monitoring of the AI's own cognitive condition**.

Another limitation is the dependence on **explicit signals of success or failure**, such as training labels, loss values, rewards, external feedback, verification mechanisms, or observable task outcomes. Current systems therefore often recognize problems only when sufficient evidence of the error is already available.

Furthermore, current AI does not generally maintain a dedicated mechanism for learning **how its internal cognitive state changes before an error occurs**. This limits its ability to recognize weak combinations of internal conflict or instability that may indicate an emerging problem.

These limitations suggest the need for a different computational approach in which an AI can continuously monitor its own internal cognitive states, develop a **pre-diagnostic warning before the exact error is known**, and use that warning to trigger deeper investigation. This gap provides the motivation for the proposed **Preconscious Error Intelligence Framework (PEIF)** introduced in Part C.

4.5. Review of Recent Research and Identified Research Gaps

To understand the current progress and remaining limitations of AI error detection, this study reviews recent research on self-monitoring, error identification, and self-correction in artificial intelligence systems. The purpose is to identify existing capabilities, examine unresolved challenges, and determine how these findings relate to the proposed Preconscious Error Intelligence Framework (PEIF).

4.5.1. Metacognitive Self-Correction for Multi-Agent Systems — Shen et al. (2026)

Research Overview

Shen et al. (2026) introduced a framework called Metacognitive Self-Correction (MASC) to improve error detection and correction in Large Language Model (LLM)-based multi-agent systems [XX]. Their research focused on a common problem in multi-agent AI: when one agent makes a mistake, that mistake can spread to other agents and affect the final decision.

Proposed Approach

The researchers developed MASC to monitor individual steps in a multi-agent system. It learns patterns of normal agent behaviour and compares new steps with those expected patterns.

The framework uses two main techniques: Next-Execution Reconstruction, which predicts the expected representation of the next step based on earlier interactions, and Prototype-Guided Enhancement, which provides a reference for normal behaviour.

When the system identifies an unusual step, it generates an anomaly score. If the score exceeds a predefined threshold, a correction agent is activated to review the output before it is passed to other agents.

Key Research Findings

The researchers evaluated MASC using the Who&When and AgentErrorBench benchmarks. Their experiments showed improvements in identifying incorrect steps compared with several existing methods.

The study reported up to an 8.47% AUC-ROC improvement in a challenging error-detection setting without access to ground-truth answers.

The researchers also demonstrated improvements in the overall performance of multi-agent systems. For example, when MASC was integrated with an LLM-Debate framework, average task accuracy increased from 87.53% to 88.89%.

These results demonstrate that monitoring individual execution steps and correcting suspicious outputs can improve the reliability of multi-agent AI systems.

Identified Limitations and Research Gaps

Although MASC achieved promising results, the researchers acknowledged important limitations.

First, its current implementation depends on a predefined threshold to decide whether an anomaly requires correction. This introduces an additional parameter that must be selected.

Second, MASC assumes access to internal communications between agents. Applying it to black-box systems, where internal interactions are not available, remains a challenge.

Furthermore, MASC primarily detects deviations in agent execution steps by comparing outputs with expected interaction patterns. This is different from the broader internal cognitive-state monitoring proposed in PEIF, which aims to combine signals such as confidence changes, reasoning inconsistencies, memory conflicts, and conflicts between alternative decisions.

Relevance to the Proposed PEIF Framework

The findings of Shen et al. provide evidence that real-time AI self-monitoring and targeted correction are technically possible and can improve system performance.

This research is closely related to PEIF because both approaches seek to detect potential problems during AI task execution rather than relying only on final-output evaluation.

However, PEIF proposes a more general conceptual architecture centred on Cognitive Tension, a warning state that would combine multiple internal signals and guide further investigation. PEIF also proposes an Error Trace Memory mechanism to learn from previous warnings, confirmed errors, and false alarms.

Therefore, the MASC study provides a valuable foundation and comparison point for PEIF. It also highlights the need for further research into reliable warning generation, access to internal cognitive information, adaptive monitoring, and effective error correction.

The effectiveness and distinct advantages of PEIF over approaches such as MASC remain to be established through future experimental evaluation.

4.5.2. A Unifying Review of Deep and Shallow Anomaly Detection — Ruff et al. (2021)

Research Overview

Ruff et al. (2021) presented a comprehensive review titled *A Unifying Review of Deep and Shallow Anomaly Detection*, published in the Proceedings of the IEEE [XX]. The study examined traditional machine learning and deep learning approaches used to identify unusual patterns or abnormal behaviour in data.

The main objective of the research was to establish a common understanding of existing anomaly-detection techniques, compare their underlying principles, evaluate their effectiveness, and identify important limitations and opportunities for future research.

Proposed Approach

The researchers introduced a unified perspective that connects traditional shallow anomaly-detection methods with modern deep learning approaches.

They examined three major categories of anomaly detection:

1. **Probabilistic Models:** Identify unusual observations based on how likely they are to occur under learned data patterns.
2. **One-Class Classification:** Learns the characteristics of normal data and identifies observations that fall outside the expected region.
3. **Reconstruction-Based Models:** Learn to reconstruct normal data and use reconstruction differences to identify potential anomalies.

The researchers also considered distance-based methods and used explainable AI techniques to investigate how anomaly-detection models make their decisions.

Key Research Findings

The study demonstrated that anomaly-detection techniques can identify abnormal patterns across different applications, including cybersecurity, healthcare, industrial monitoring, and fraud detection.

However, the researchers found that no single anomaly-detection method consistently performed best across all tested conditions.

Their experimental comparisons using the MNIST-C and MVTec-AD datasets showed that the effectiveness of different models depends on the characteristics of the data and the type of anomaly being detected.

An important finding involved an industrial anomaly-detection experiment using wood images. The researchers observed that a model could achieve strong detection performance while relying on irrelevant image features rather than the actual cause of the anomaly.

This demonstrated that detecting an unusual pattern does not necessarily mean the system has correctly understood why the observation is abnormal.

The study also highlighted the importance of explainability, appropriate evaluation methods, and reliable identification of abnormal patterns.

Identified Limitations and Research Gaps

The researchers identified several challenges associated with existing anomaly-detection methods.

First, distinguishing normal behaviour from abnormal behaviour is difficult when normal data contain significant variations. This can produce false warnings or cause genuine anomalies to remain undetected.

Second, anomaly-detection performance can decrease when the characteristics of incoming data change over time or differ from the training data.

Third, some models can successfully identify anomalies while relying on misleading features or unrelated patterns. Therefore, high detection accuracy does not necessarily guarantee that a model has correctly identified the underlying cause.

Additionally, the researchers highlighted challenges involving model interpretation, computational complexity, and the reliability of anomaly-detection systems in real-world environments.

These findings demonstrate that anomaly detection remains a challenging research problem requiring more reliable, explainable, and adaptable approaches.

Relevance to the Proposed PEIF Framework

The findings of Ruff et al. are relevant to the Preconscious Error Intelligence Framework (PEIF) because they demonstrate both the capabilities and limitations of conventional anomaly-detection approaches.

Traditional anomaly detection generally focuses on identifying deviations from learned normal patterns within the data or observations being monitored. Such techniques can also be applied to internal system information when suitable signals are available.

However, the reviewed paper does not specifically investigate a unified cognitive architecture that continuously combines multiple internal reasoning signals to produce a pre-diagnostic warning state.

PEIF proposes a broader monitoring process that considers possible inconsistencies in an AI system's internal reasoning, including confidence changes, expectation mismatches, memory conflicts, and contradictory intermediate results.

Instead of treating an unusual signal as confirmation of an error, PEIF proposes generating a warning state called Cognitive Tension. This state would indicate that further investigation may be necessary before the exact cause of the problem is identified.

Furthermore, PEIF proposes an adaptive learning mechanism in which previous warnings, confirmed errors, successful corrections, and false alarms are recorded to improve future monitoring.

Therefore, the research by Ruff et al. provides important theoretical foundations for understanding anomaly detection and its limitations. It also offers established techniques that could serve as baseline methods when evaluating PEIF.

The proposed advantages of PEIF over existing anomaly-detection approaches would need to be demonstrated through future experimental comparisons.

4.5.3. Large Language Models Cannot Self-Correct Reasoning Yet — Huang et al. (2024)

Research Overview

Huang et al. (2024) conducted a study titled *Large Language Models Cannot Self-Correct Reasoning Yet*, published at the International Conference on Learning Representations (ICLR 2024) [XX]. The research examined whether Large Language Models (LLMs) can independently identify and correct mistakes in their own reasoning without receiving external feedback.

The researchers focused on a concept called intrinsic self-correction, where an AI model attempts to review and improve its previous answer using only its existing knowledge and reasoning capabilities.

The main objective was to determine whether LLMs genuinely possess reliable self-correction abilities or whether improvements observed in previous studies depend on external feedback, additional prompts, or experimental conditions.

Proposed Approach

The researchers evaluated several language models, including GPT-3.5, GPT-4, GPT-4-Turbo, and Llama-2, using reasoning benchmarks such as GSM8K, CommonSenseQA, and HotpotQA.

The experiments followed a self-correction process consisting of three main stages:

1. **Initial Response Generation:** The AI generates an answer to a reasoning problem.
2. **Self-Reflection:** The AI is prompted to review its previous answer and identify possible mistakes.
3. **Self-Correction:** The AI attempts to improve or correct its original answer based on its own feedback.

The researchers compared two conditions: self-correction supported by external information about answer correctness and intrinsic self-correction without such information.

They also investigated whether multi-agent debate and different prompt designs genuinely improved reasoning performance.

Key Research Findings

The results demonstrated that the language models tested struggled to correct their own reasoning errors reliably without external feedback.

For example, on the GSM8K mathematical reasoning benchmark, GPT-4 achieved an initial accuracy of 95.5%. However, after one round of intrinsic self-correction, accuracy decreased to 91.5%, and after two rounds, it decreased further to 89.0%.

Similarly, GPT-3.5 achieved an initial accuracy of 75.9% on GSM8K, which decreased to 74.7% after two rounds of intrinsic self-correction.

The researchers also found that models sometimes changed initially correct answers into incorrect answers during self-correction.

In contrast, self-correction guided by reliable external information about answer correctness produced improvements. For example, GPT-3.5 accuracy on GSM8K increased from 75.9% to 84.3% when such feedback was available.

The experiments also showed that multi-agent debate did not outperform simpler self-consistency methods when the number of generated responses was taken into account.

These findings suggest that repeatedly asking an LLM to review its own answer does not necessarily improve reasoning accuracy.

Identified Limitations and Research Gaps

The research highlighted several important limitations in existing LLM self-correction methods.

First, the tested LLMs showed difficulty reliably determining whether their own reasoning was correct without external feedback.

Second, additional self-reflection prompts sometimes caused models to change correct answers into incorrect ones, reducing overall performance.

Third, some self-correction approaches depended heavily on the quality of external feedback or the design of prompts. Improvements reported in earlier studies did not always represent genuine independent error identification.

Furthermore, the study demonstrated that using additional model responses and repeated reasoning steps could increase computational costs without necessarily improving accuracy.

The research primarily examined reasoning tasks using the selected language models and self-correction strategies. Therefore, its findings should not be interpreted as proof that every present or future AI system is incapable of independent error detection.

Nevertheless, the results highlight an important research challenge: developing reliable mechanisms that can identify potential reasoning failures and determine when additional verification is necessary.

Relevance to the Proposed PEIF Framework

The findings of Huang et al. are highly relevant to the proposed Preconscious Error Intelligence Framework (PEIF), particularly its emphasis on identifying possible reasoning problems before attempting detailed correction.

The research demonstrates that simply prompting an AI model to reconsider its previous answer may be insufficient for reliable self-correction.

To explore an alternative approach, PEIF proposes a dedicated Internal Monitoring Module that would continuously examine multiple available reasoning-related signals, such as confidence changes, inconsistencies between intermediate results, conflicting evidence, and deviations from expected behaviour.

Instead of immediately attempting to correct an answer, PEIF proposes generating a pre-diagnostic warning state called Cognitive Tension when potentially significant internal inconsistencies are detected.

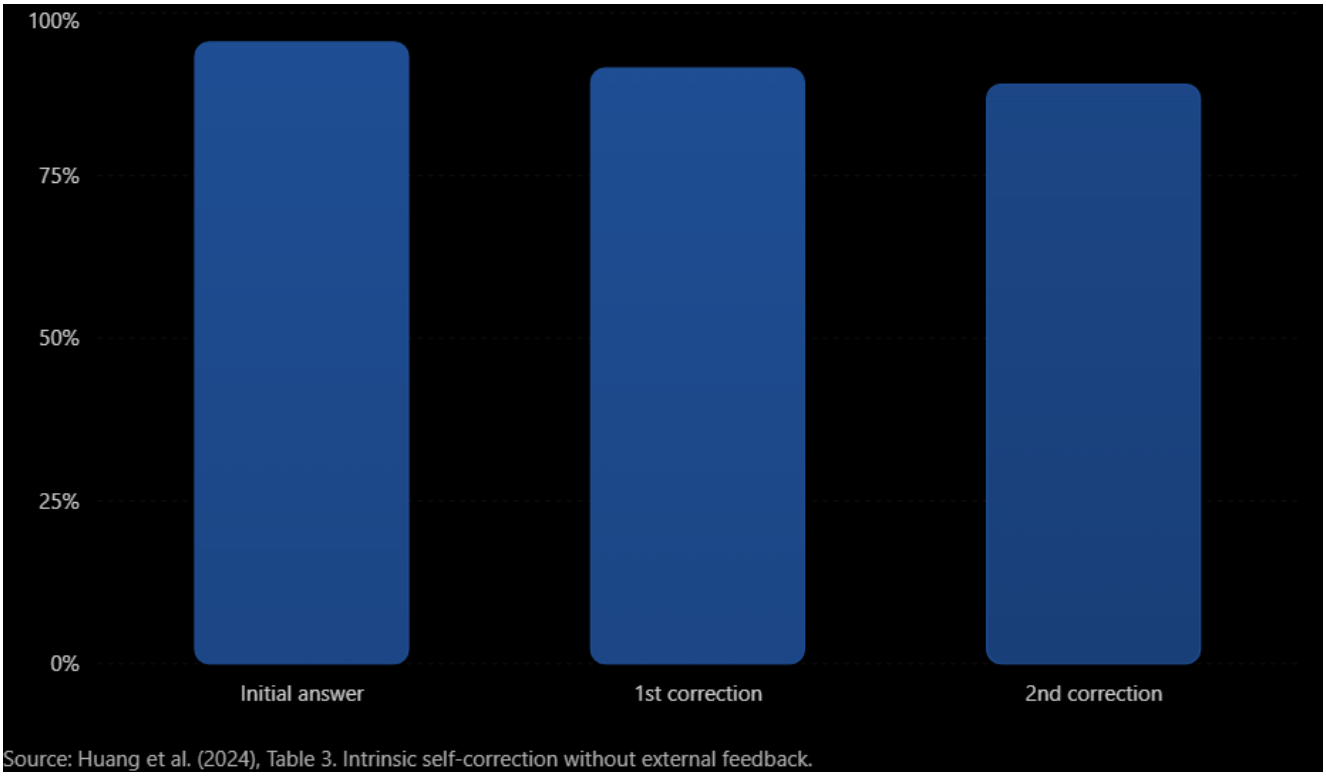
This warning would activate an Attention and Diagnostic Reasoning Module, allowing the system to investigate suspicious reasoning steps before deciding whether to continue, verify, or correct its response.

PEIF also proposes a Memory and Adaptation Module that would use previous warnings, confirmed errors, false alarms, and successful corrections to improve future monitoring.

Therefore, the study by Huang et al. provides empirical motivation for investigating structured error-monitoring mechanisms that go beyond repeated self-reflection prompting.

However, PEIF remains a conceptual framework. Further experiments are required to determine whether its proposed monitoring approach can reliably identify reasoning errors, avoid unnecessary corrections, and achieve better results than existing self-correction techniques.

GPT-4 accuracy on GSM8K mathematical reasoning



Source: Huang et al. (2024), Table 3. Intrinsic self-correction without external feedback.

4.5.4. Predictive Metacognition: A Neuro-Computational Framework for Self-Monitoring in Large Language Models — Luo and Jho (2026)

Research Overview

Luo and Jho (2026) introduced a framework called *Predictive Metacognition*, published in *Scientific Reports* [XX]. Their research investigated how human brain-inspired monitoring mechanisms could improve the ability of Large Language Models (LLMs) to evaluate the reliability of their own responses.

The researchers identified overconfidence, unreliable uncertainty estimation, and hallucination as important limitations of existing language models. Although these models can generate convincing responses, their expressed confidence does not always reflect whether their answers are correct.

The main objective of the study was to develop an internal self-monitoring mechanism that enables language models to generate responses while simultaneously estimating the reliability of those responses.

Proposed Approach

The researchers developed a neuro-inspired framework based on predictive processing and the error-monitoring functions associated with the human anterior cingulate cortex (ACC).

Their architecture introduced two parallel processing streams:

1. **Task-Oriented Processing:** Generates responses to user questions or tasks.
2. **Metacognitive Monitoring:** Examines internal computational information to estimate the likelihood that the generated response is correct.

The proposed approach incorporated four principles: hierarchical prediction, error-driven learning, attention-based confidence estimation, and dual-process monitoring.

Instead of relying only on prompts asking the model to check its answer after generation, the framework trained language models to produce confidence estimates alongside their normal responses.

The researchers implemented this approach using Llama-3-8B-Instruct and Phi-3-Mini-4k-Instruct. They applied Low-Rank Adaptation (LoRA) fine-tuning using 4,000 training examples and a combined learning objective that considered both language generation and confidence prediction.

The framework also used a two-stage confidence assessment. When an initial confidence estimate fell below a selected threshold, an additional prediction mechanism produced a more detailed confidence assessment.

Key Research Findings

The researchers reported measurable improvements in the ability of the fine-tuned models to estimate their own response reliability.

For Llama-3-8B-Instruct, the Brier Score decreased from 0.138 to 0.122, representing approximately an 11.6% improvement. Its Expected Calibration Error (ECE) also decreased from 0.252 to 0.224.

Similarly, Phi-3-Mini-4k-Instruct achieved a reduction in Brier Score from 0.155 to 0.128, while its ECE decreased from 0.274 to 0.231.

The researchers also reported improvements in task accuracy. The fine-tuned Llama model increased from 83.7% to 85.3%, while the Phi model increased from 82.0% to 84.7%.

Additional evaluations suggested that the learned confidence-estimation capability could transfer to tasks outside the training domain, although calibration performance still changed under domain shifts.

These findings provide experimental evidence that brain-inspired monitoring techniques can improve some aspects of an AI system's ability to assess its own reliability.

Identified Limitations and Research Gaps

Although the framework demonstrated promising improvements, the researchers identified several remaining limitations.

First, the framework primarily focused on predicting response correctness and improving confidence calibration. Its evaluation did not establish a complete process for identifying the exact source of a reasoning error and performing targeted correction.

Second, the implemented training process relied on binary correctness labels, distinguishing between correct and incorrect responses. This simplifies the complexity of real reasoning errors, which may involve partially correct answers, conflicting evidence, or several uncertain reasoning steps.

Third, the implementation provided explicit metacognitive supervision mainly at the final response level rather than fully developed monitoring across multiple reasoning levels.

Fourth, the researchers evaluated models with up to eight billion parameters. Further experiments are required to establish whether the approach scales effectively to larger language models and more complex autonomous AI systems.

Finally, the framework was primarily evaluated through confidence estimation and response reliability. Additional research is needed to investigate how monitoring signals can guide detailed error diagnosis, adaptive decision-making, and learning from previous error events during ongoing operation.

Relevance to the Proposed PEIF Framework

The research by Luo and Jho is highly relevant to the proposed Preconscious Error Intelligence Framework (PEIF), as both approaches draw inspiration from human cognitive monitoring and introduce processes that operate alongside ordinary AI task execution.

Importantly, their work demonstrates that parallel internal monitoring is not merely a theoretical possibility. It can be implemented computationally and can produce measurable improvements in confidence calibration.

However, the primary focus of Predictive Metacognition is estimating the probability that an AI-generated response is correct.

PEIF proposes a broader conceptual process centred on pre-diagnostic error awareness. Rather than relying only on a confidence estimate, PEIF aims to combine different warning signals, including confidence changes, reasoning conflicts, unexpected intermediate results, and inconsistencies between available evidence.

These signals would contribute to a proposed Cognitive Tension state, representing the possibility that further investigation is required before the exact problem has been identified.

PEIF also introduces an Attention and Diagnostic Reasoning Module intended to investigate the suspicious reasoning process and identify possible causes. Based on the investigation, the system would decide whether to continue, verify, or correct its response.

Additionally, PEIF proposes an Error Trace Memory and adaptation mechanism that would retain information about previous warnings, confirmed errors, false alarms, and successful corrections to support future monitoring decisions.

Therefore, the Predictive Metacognition study provides both scientific support and practical implementation guidance for PEIF. It also establishes an important existing approach against which PEIF should be compared.

The proposed distinction of PEIF lies in combining pre-diagnostic warning generation, triggered reasoning investigation, adaptive response selection, and experience-based monitoring within one conceptual architecture. These features remain proposed contributions, and their effectiveness and originality relative to existing systems require further implementation and comparative evaluation.

5. Part C – Designing the Next Generation AI Framework

The proposed **Preconscious Error Intelligence Framework (PEIF)** is a conceptual, brain-inspired Artificial Intelligence architecture designed to model a functional equivalent of human unconscious error detection. Its underlying philosophy is that future AI systems should not only evaluate the correctness of their outputs but also monitor the stability and coherence of their own cognitive processing.

The central principle of PEIF is **"Knowing THAT something may be wrong before knowing WHAT is wrong."** Rather than immediately identifying an error, the framework aims to recognize early signs of internal inconsistency and use them to initiate further investigation.

PEIF introduces a parallel cognitive monitoring process that observes internal states such as prediction instability, uncertainty changes, memory conflicts, goal inconsistencies, and reasoning conflicts. These signals are integrated into a pre-diagnostic warning state called **Cognitive Tension**. This proposed mechanism is inspired by early human error-monitoring processes without assuming that AI possesses consciousness or subjective awareness.

5.1. Core Computational Principles

PEIF is based on five interconnected computational principles:

1. **Continuous Internal Monitoring:** The system observes accessible internal cognitive states while the primary AI performs reasoning, planning, prediction, and decision-making.
2. **Distributed Inconsistency Detection:** Multiple weak signals, including expectation mismatches, uncertainty changes, and internal conflicts, are evaluated collectively rather than relying on a single error indicator.
3. **Cognitive Tension Generation:** The framework integrates these signals into a dynamic warning state representing possible cognitive instability before an exact error is identified.
4. **Adaptive Attention Allocation:** Significant Cognitive Tension triggers additional diagnostic reasoning, while low tension allows normal processing to continue.
5. **Error Trace Learning:** Internal patterns associated with previous failures and false warnings are stored and used to improve future monitoring and adaptation.

Together, these principles establish a proposed separation between ordinary task processing, early error sensing, and deliberate error diagnosis.

5.2. Major Architectural Components

The proposed framework consists of the following interconnected components:

Primary Cognitive Workspace: Performs normal AI operations, including perception, reasoning, planning, prediction, and decision-making. It also generates internal cognitive-state information that can be monitored.

Internal Monitoring and Expectation Module: Observes accessible internal signals and compares current cognitive states with expected patterns. It identifies potential inconsistencies involving memory, goals, predictions, confidence, and reasoning.

Preconscious Integration and Cognitive Tension Module: Combines multiple weak inconsistency signals into a graded Cognitive Tension state. This state indicates that further investigation may be necessary without assuming that an actual error has occurred.

Adaptive Attention Controller: Determines whether additional computational resources should be allocated based on Cognitive Tension, task importance, available resources, and potential risk.

Diagnostic Reasoning Module: Investigates the possible cause of a warning. If an error is identified, the system attempts correction. If the cause remains unknown, it may explore alternative explanations, retrieve additional information, or request human assistance.

Memory and Adaptation Module: Maintains task knowledge, working information, and Error Trace Memory. It supports learning from previous warnings, identified errors, successful corrections, and false alarms.

These components form an integrated architecture in which internal monitoring operates alongside the primary cognitive process.

5.3. Learning and Memory Mechanism

PEIF proposes an experience-based learning process called **Error Trace Consolidation**. When the system generates a warning, it records the associated internal cognitive-state pattern and the eventual outcome of diagnostic investigation.

If a warning is associated with a confirmed error, the relevant internal pattern may receive greater importance during future monitoring. If the warning is repeatedly found to be unnecessary, its influence may be reduced. This allows the framework to adjust its sensitivity based on experience.

The memory architecture contains three main forms of memory:

- **Knowledge Memory:** Stores task-related information, learned concepts, and relevant expectations.

- **Working Memory:** Maintains information associated with the system's current reasoning and decision-making.
- **Error Trace Memory:** Stores internal-state patterns associated with previous errors, warnings, and recovery outcomes.

Through these mechanisms, PEIF aims to learn not only from completed failures but also from the internal conditions that preceded them.

5.4. Reasoning and Decision Making

PEIF separates cognitive processing into three functional levels.

The **Primary Cognitive Level** performs the main task, such as generating an answer, making a prediction, or planning an action.

The **Preconscious Monitoring Level** evaluates internal cognitive stability and generates Cognitive Tension when unusual combinations of internal signals are detected.

The **Diagnostic Cognitive Level** investigates significant warnings to determine whether an error exists and, where possible, identify its cause.

Decision-making is controlled by the Adaptive Attention Controller. When Cognitive Tension remains low, normal processing continues. When tension exceeds an adaptive threshold, the system may pause, verify information, reconsider its reasoning, change its strategy, or request additional evidence.

Importantly, a high Cognitive Tension value does not guarantee an error. It represents a reason to investigate. This distinction prevents the framework from treating every internal inconsistency as a confirmed failure.

5.5. Adaptation and Evolution

PEIF is designed to adapt through repeated monitoring, diagnosis, and experience learning. Its monitoring sensitivity can be adjusted based on confirmed errors, missed warnings, false alarms, and successful corrections.

The Expectation Model can evolve as the system encounters new tasks and environments, while Error Trace Memory can accumulate patterns associated with previously observed failures.

Over time, these updates could allow PEIF to improve its ability to recognize emerging cognitive instability. However, the effectiveness of this adaptation would require experimental validation.

5.6. Expected Intelligent Behaviours

If successfully implemented, PEIF is expected to demonstrate several intelligent behaviours:

1. Recognizing possible internal problems before their exact causes are identified.
2. Allocating additional attention when cognitive instability becomes significant.
3. Investigating inconsistencies without automatically assuming that an error exists.
4. Exploring alternative explanations when the cause of a warning remains unknown.
5. Learning internal warning patterns from previous failures and successful recoveries.
6. Adjusting monitoring sensitivity according to task risk and accumulated experience.

These behaviours represent proposed functional capabilities rather than evidence of machine consciousness.

5.7. Proposed Advantages Over Existing AI Paradigms

Unlike approaches that rely primarily on output verification or externally observed failures, PEIF proposes an integrated process for monitoring internal cognitive-state changes during task execution.

Its potential advantages include earlier identification of situations requiring verification, adaptive allocation of diagnostic resources, improved handling of uncertain error causes, and experience-based refinement of monitoring behaviour.

Although existing AI research already includes anomaly detection, uncertainty estimation, self-monitoring, and metacognitive mechanisms, PEIF proposes to integrate these capabilities around a dedicated pre-diagnostic Cognitive Tension state.

The originality of PEIF therefore lies in its proposed architectural organization and interactions rather than the invention of every individual monitoring technique. Its practical advantages must be established through future comparative experiments.

5.7.1. A Brain-Inspired Agentic Architecture to Improve Planning with LLMs — Webb et al. (2025)

Research Overview

Webb et al. (2025) presented a research study titled *A Brain-Inspired Agentic Architecture to Improve Planning with LLMs*, published in *Nature Communications* [XX]. The study investigated how computational principles inspired by the human prefrontal cortex could be applied to improve planning and reasoning in Large Language Models (LLMs).

The researchers identified an important limitation in existing language models: although LLMs can perform individual reasoning tasks, they often struggle to coordinate multiple reasoning processes when solving complex problems. This can lead to invalid actions, incorrect decisions, and failures during multi-step planning.

The main objective was to investigate whether separating different cognitive functions into specialised AI modules could improve planning performance and reduce errors.

Proposed Approach

The researchers introduced a framework called the Modular Agentic Planner (MAP), which consists of multiple specialised LLM-based modules inspired by functions associated with different regions of the human prefrontal cortex.

The architecture includes six main components:

1. **Task Decomposer:** Divides complex goals into smaller tasks.
2. **Actor:** Proposes possible actions to achieve a given goal.
3. **Monitor:** Checks whether proposed actions violate task constraints and provides feedback when invalid actions are identified.
4. **Predictor:** Predicts the possible outcomes of proposed actions.
5. **Evaluator:** Estimates the usefulness of predicted outcomes.
6. **Orchestrator:** Coordinates the planning process and determines when goals have been achieved.

The Monitor module is particularly relevant to the present research because it was inspired by the Anterior Cingulate Cortex (ACC), a human brain region associated with conflict monitoring and error processing.

In MAP, the Monitor examines proposed actions and checks whether they violate the rules of the current task. When an invalid action is detected, the system provides feedback to the Actor, allowing it to propose an alternative.

This demonstrates how a specialised monitoring process can support the identification and prevention of certain errors during AI decision-making.

Key Research Findings

The researchers evaluated MAP on several planning and reasoning benchmarks, including Tower of Hanoi, graph traversal tasks, PlanBench, and StrategyQA.

The experiments demonstrated that MAP could improve performance compared with several standard LLM prompting techniques and agent-based approaches.

One particularly important finding was reported in the Tower of Hanoi experiment. The complete MAP architecture solved an average of 74% of the tested three-disk problems, while a version without the Monitor module solved only 27%.

Furthermore, the complete MAP architecture produced no invalid actions in this experiment, whereas removing the Monitor resulted in 31% invalid actions.

These findings demonstrate that a dedicated monitoring module can make a substantial contribution to an AI system's ability to detect and prevent invalid actions.

The researchers also found that MAP improved performance on several unfamiliar planning tasks, although its generalisation remained imperfect.

Overall, the study provides experimental evidence that separating planning and monitoring functions into specialised modules can improve the reliability of LLM-based systems.

Identified Limitations and Research Gaps

Despite its promising results, the study identified several limitations.

First, the MAP framework primarily focuses on goal-directed planning and checking proposed actions against task constraints. Its Monitor determines whether an action is valid or invalid according to available rules.

However, this is different from identifying a broader internal warning state based on uncertain or conflicting reasoning signals before the exact error has been diagnosed.

Second, the Monitor produces a validity assessment and feedback for proposed actions. The study does not establish a general mechanism that combines multiple internal cognitive signals into a unified pre-diagnostic warning state.

Third, the proposed architecture was mainly evaluated in fully observable and deterministic environments. Further research is required to examine its effectiveness in uncertain, dynamic, and open-ended situations.

Fourth, using multiple specialised LLM modules introduces additional computational costs compared with simpler approaches.

Finally, the researchers identified opportunities for incorporating more advanced memory mechanisms, improving generalisation, and extending the architecture to more complex reasoning environments.

These limitations suggest opportunities for investigating broader and more adaptive AI monitoring mechanisms.

Relevance to the Proposed PEIF Framework

The research by Webb et al. provides valuable scientific and computational support for the proposed Preconscious Error Intelligence Framework (PEIF).

Both MAP and PEIF draw inspiration from the human brain and propose separating ordinary AI processing from specialised monitoring functions.

MAP demonstrates that an ACC-inspired monitoring module can be implemented and can reduce invalid actions during AI planning.

However, PEIF proposes extending this direction toward a broader form of pre-diagnostic error monitoring.

Instead of evaluating only whether proposed actions violate known rules, PEIF aims to monitor multiple available signals, including reasoning inconsistencies, confidence changes, unexpected intermediate results, and conflicting evidence.

These signals would contribute to a proposed warning state called Cognitive Tension, which represents the possibility that something may be wrong before the system has identified the exact cause.

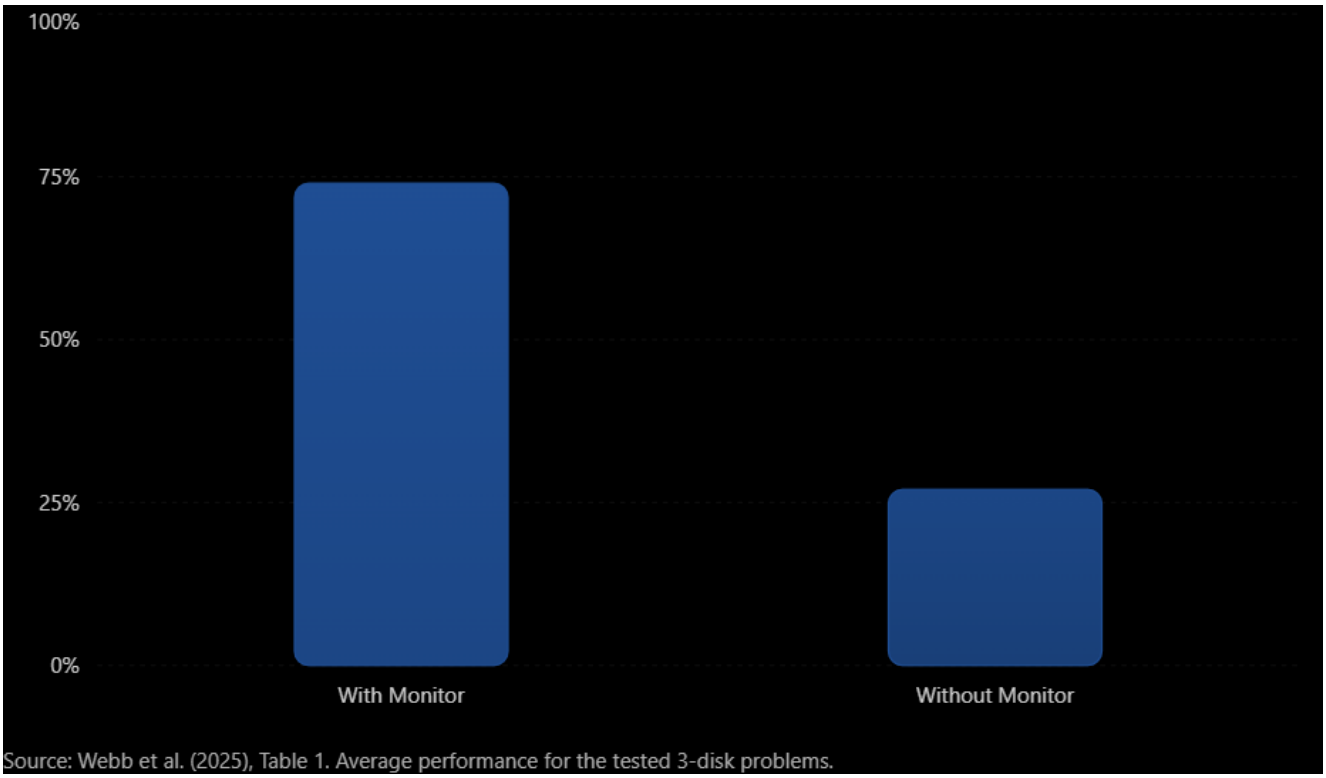
When Cognitive Tension indicates a significant potential problem, PEIF proposes activating an Attention and Diagnostic Reasoning Module to investigate the suspicious reasoning process.

The system would then decide whether to continue, verify, or correct its reasoning, while a Memory and Adaptation Module would retain previous warning outcomes to support future improvements.

Therefore, Webb et al.'s research provides practical evidence supporting the feasibility of modular, brain-inspired monitoring architectures.

It also establishes an important existing approach against which PEIF should be compared. While MAP focuses on planning and action validation, PEIF proposes a more general architecture connecting multi-signal early warnings, targeted diagnosis, adaptive responses, and experience-based monitoring.

These proposed differences do not establish that PEIF performs better than MAP. Further implementation and experimental evaluation are required to determine whether PEIF provides measurable advantages over existing brain-inspired AI architectures.



Source: Webb et al. (2025), Table 1. Average performance for the tested 3-disk problems.

With Monitor

0%

Invalid actions

Without Monitor

31%

Invalid actions

These are the researchers' results, not PEIF results. They show the benefit of monitoring task-rule violations within MAP.

5.7.2. How Metacognitive Architectures Remember Their Own Thoughts: A Systematic Scoping Review — Nolte et al. (2026)

Research Overview

Nolte et al. (2026) conducted a systematic scoping review titled *How Metacognitive Architectures Remember Their Own Thoughts*, published in *Artificial Intelligence Review* [XX]. The study investigated how artificial intelligence systems can monitor, represent, store, and reuse information about their own cognitive processes.

The research focused on Computational Metacognitive Architectures (CMAs), which are AI systems designed to examine and regulate aspects of their own information processing.

The researchers identified that existing metacognitive architectures use different methods for representing and remembering internal processing experiences. However, inconsistent terminology, limited evaluation, and differences between implementations make it difficult to compare their effectiveness.

The main objective was to systematically examine existing approaches to metacognitive memory and identify opportunities for developing more reliable, adaptable, and explainable AI systems.

Research Methodology and Approach

The researchers conducted a systematic literature search across 16 academic databases and reviewed 101 research reports covering 35 distinct Computational Metacognitive Architectures.

The review examined three major aspects of these systems:

1. **Architectural Design:** How AI systems separate normal cognitive processing from higher-level monitoring and control.
2. **Memory Representation:** How systems record internal events, reasoning steps, confidence information, errors, and previous decisions.
3. **Processing and Evaluation:** How stored information is used for detecting problems, diagnosing failures, supporting correction, and improving future behaviour.

The researchers identified several approaches to metacognitive monitoring, including predefined pattern matching, machine learning-based anomaly detection, continuous adaptation, and reasoning based on previous experiences.

The study also examined existing architectures such as Meta-AQUA, MIDCA, MCL, CARINA, and Soar, which use different combinations of monitoring, memory, diagnosis, and adaptation.

Key Research Findings

The review identified several important findings.

First, the researchers found that metacognitive memory can support AI systems in recognising failures, improving decisions, explaining their behaviour, and learning from previous experiences.

Second, many existing architectures separate ordinary task processing from a higher-level monitoring process. This demonstrates that cognitive and metacognitive functions can be organised into interacting computational components.

Third, the study identified a common metacognitive processing cycle involving monitoring, modelling, diagnosis, identification of failure causes, strategy selection, and repair.

Fourth, the researchers found that different systems store internal processing information in different formats, including reasoning traces, episodic memories, event records, and symbolic representations.

An important finding was the limited experimental evaluation of existing approaches. Only approximately 17% of the reviewed architectures had quantitative comparisons that specifically tested the contribution of their metacognitive experience-processing mechanisms.

The researchers also reported substantial variation in terminology, implementation details, and evaluation methods.

Overall, the findings demonstrate that AI self-monitoring and memory-based adaptation are established research directions, but the evidence supporting their effectiveness remains uneven.

Identified Limitations and Research Gaps

The review highlighted several limitations in existing computational metacognitive architectures.

First, the lack of standardised representations and terminology makes it difficult to compare monitoring and memory mechanisms across different systems.

Second, many architectures have been evaluated in relatively simple or restricted environments. Their effectiveness in complex, uncertain, and dynamic situations remains insufficiently established.

Third, only a limited number of studies have quantitatively isolated the contribution of metacognitive memory. Consequently, the specific benefits of remembering internal processing experiences remain difficult to evaluate consistently.

Fourth, some architectures provide insufficient details about the algorithms and data structures used for storing and processing internal reasoning information.

Fifth, integrating structured metacognitive information with modern neural networks and Large Language Models remains challenging because their internal processing is not always directly interpretable.

The authors also acknowledged limitations in their own review, including incomplete information in the original studies and the possible under-representation of approaches based entirely on neural networks.

These findings highlight the need for more clearly specified, experimentally evaluated, and adaptable metacognitive systems.

Relevance to the Proposed PEIF Framework

The research by Nolte et al. provides an important theoretical and architectural foundation for the Preconscious Error Intelligence Framework (PEIF).

PEIF proposes that an AI system should continuously monitor available internal signals, recognise potential reasoning problems, and generate an early warning before identifying the exact cause.

The review demonstrates that monitoring internal processes and maintaining records of previous cognitive experiences are computationally feasible approaches that have already been explored in existing AI architectures.

These findings support the conceptual design of PEIF's Internal Monitoring Module and Memory and Adaptation Module.

In PEIF, the Internal Monitoring Module would examine multiple available reasoning-related signals, including confidence changes, conflicting information, unexpected intermediate results, and deviations from expected behaviour.

These signals would contribute to a proposed Cognitive Tension state, indicating that a possible problem requires further investigation.

The Memory and Adaptation Module would record relevant information about previous warnings, identified errors, false alarms, successful corrections, and the circumstances in which these events occurred.

This information could then be used to investigate recurring problems and improve future monitoring decisions.

Importantly, the reviewed literature already includes architectures that combine monitoring, anomaly detection, diagnosis, and adaptive correction. Therefore, PEIF does not claim to introduce these individual capabilities for the first time.

Instead, PEIF proposes a specific integration of multi-signal warning generation, pre-diagnostic Cognitive Tension, targeted investigation, adaptive response selection, and experience-based monitoring.

The review also provides guidance for future evaluation. PEIF should be compared with existing monitoring architectures using measurable outcomes such as warning accuracy, false-alarm rates, error-detection time, correction effectiveness, and computational overhead.

Therefore, Nolte et al.'s study provides a valuable scientific foundation for PEIF's monitoring and memory components while highlighting the need for clear implementation details and rigorous experimental validation.

The proposed effectiveness of PEIF remains to be demonstrated through future research.

Existing MIDCA approach

D-Track

Detects possible anomalies using learned patterns

K-Track

Investigates problems in greater detail using cognitive traces

Our proposed PEIF approach

Internal Monitoring + Cognitive Tension

Combines proposed warning signals into an early-warning state

Attention and Diagnostic Reasoning

Investigates potential problems before choosing a response

Scientific Foundation of Memory-Based Adaptation

The proposed Memory and Adaptation Module is supported by research on computational metacognition and episodic memory. Nolte et al. (2026) systematically reviewed 35 Computational Metacognitive Architectures and demonstrated that several existing systems maintain records of their internal processing to support self-monitoring, error diagnosis, and adaptation [XX].

Their findings suggest that preserving relevant information about previous cognitive processes can help AI systems analyse past failures and guide future behaviour. Building on this established principle, PEIF proposes maintaining structured records of pre-diagnostic warnings, monitoring signals, confirmed errors, false alarms, and correction outcomes. These records would support the refinement of future warning decisions and the identification of recurring reasoning problems.

6. Part D – Conceptual System Architecture

The conceptual architecture of the **Preconscious Error Intelligence Framework (PEIF)** illustrates how an AI system could monitor its internal cognitive processing, generate an early warning, investigate potential errors, and adapt through experience.

Unlike a purely sequential error-checking process, PEIF proposes two interacting information flows: a **primary cognitive process** responsible for normal task execution and a **parallel monitoring process** responsible for detecting possible cognitive instability.

6.1. Major System Components

The architecture begins with the **Input and Context Module**, which receives information such as text, images, sensor data, environmental conditions, and task instructions.

The **Perception and Representation Module** converts incoming information into structured representations suitable for further processing.

The **Active Cognition Module** performs reasoning, planning, prediction, and decision-making while producing accessible internal cognitive-state information.

The **Internal Monitoring Module** evaluates indicators such as uncertainty, expectation mismatch, memory conflict, goal inconsistency, and reasoning instability. An associated **Expectation Model** provides reference patterns for identifying deviations from expected cognitive behaviour.

The **Cognitive Tension Module** integrates these signals into a graded pre-diagnostic warning state. This warning is then interpreted by the **Adaptive Attention Controller**, which determines whether deeper investigation is necessary.

When investigation is triggered, the **Diagnostic Reasoning Module** attempts to identify the source of the inconsistency. The resulting response is handled through correction or further exploration.

Finally, the **Memory and Adaptation Module** records relevant internal patterns and outcomes to improve future monitoring.

6.2. Information Flow and Processing Stages

The proposed operational workflow follows eight main stages:

1. **Input Acquisition:** The system receives task information and environmental context.
2. **Active Cognitive Processing:** The primary AI interprets information and performs reasoning or decision-making.
3. **Internal State Monitoring:** The monitoring process observes accessible cognitive signals and compares them with expected patterns.

4. **Cognitive Tension Generation:** Multiple inconsistency signals are integrated into a graded warning state.
5. **Attention Allocation:** Low tension allows normal processing to continue, while significant tension triggers additional investigation.
6. **Diagnostic Reasoning:** The system investigates whether an error exists and attempts to identify its cause.
7. **Adaptive Response:** Identified errors may be corrected, while unresolved warnings may lead to exploration, additional evidence gathering, or human review.
8. **Experience Learning:** The system updates relevant memory, expectations, and monitoring sensitivity based on the outcome.

This workflow distinguishes **early error sensing** from **explicit error identification**, which is the central cognitive principle of PEIF.

6.3. Interactions Between Modules

The modules operate through continuous information exchange rather than as isolated components.

The Active Cognition Module provides internal cognitive-state information to the Internal Monitoring Module. The Expectation Model supports monitoring by representing expected cognitive patterns.

The Cognitive Tension Module combines monitoring signals and communicates the resulting warning level to the Adaptive Attention Controller.

When additional investigation is required, the controller activates the Diagnostic Reasoning Module, which may request further information from the primary cognitive process or external resources.

The Memory and Adaptation Module receives information about warnings, diagnostic outcomes, and corrections. It then updates Error Trace Memory, expected patterns, and monitoring sensitivity.

These feedback interactions allow the proposed framework to refine its future responses based on experience.

6.4. Overall Operational Workflow

Consider an AI system generating a recommendation. During reasoning, its monitoring process detects a combination of confidence instability and conflict with previously stored information.

Although the system has not identified the exact problem, these signals increase Cognitive Tension. The Adaptive Attention Controller then activates additional verification.

If the Diagnostic Reasoning Module identifies an incorrect assumption, the system revises its recommendation. If the cause remains unknown, it may retrieve additional information or request clarification rather than presenting an unsupported conclusion.

The resulting warning, investigation, and outcome are recorded by the Memory and Adaptation Module for future learning.

This example illustrates how PEIF could move from **normal cognition to early warning, targeted diagnosis, adaptive response, and experience-based learning**.

The conceptual architecture therefore represents a proposed general-purpose cognitive monitoring framework that could be integrated into different AI systems without being restricted to a specific application domain.

Figure 1: Conceptual Architecture of the Preconscious Error Intelligence Framework (PEIF)

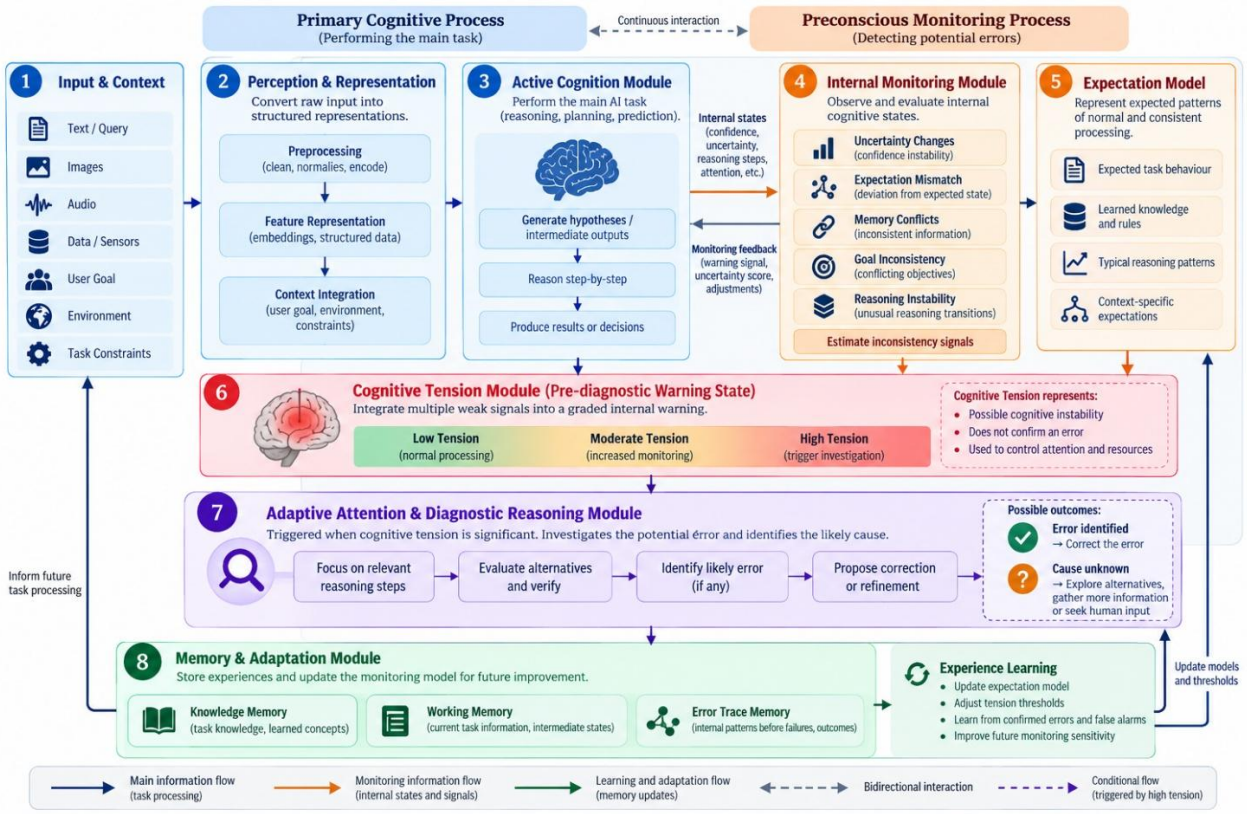


Figure 1: Conceptual Architecture of the Preconscious Error Intelligence Framework (PEIF)

7. Part E – Feasibility Discussion

The **Preconscious Error Intelligence Framework (PEIF)** is a conceptual architecture rather than an experimentally implemented system. However, several of its proposed components could be explored using existing programming languages, machine learning techniques, and computing platforms.

7.1. Feasibility of Implementation Using Python

Python could be used to develop an initial PEIF prototype because it supports machine learning, numerical computation, data processing, and modular AI system development.

A prototype could combine an existing AI model with a separate monitoring mechanism that collects accessible internal signals, estimates Cognitive Tension, and triggers additional verification when necessary.

Python would also support experimental evaluation of warning accuracy, false alarms, diagnostic effectiveness, and computational overhead.

However, Python alone would not automatically provide unconscious error detection. The central challenge would be designing and validating the proposed monitoring and Cognitive Tension mechanisms.

7.2. Suitable Python Libraries and Frameworks

Several existing Python libraries could support experimental development.

PyTorch and TensorFlow could be used to build predictive models, uncertainty estimators, and learned monitoring mechanisms.

NumPy and SciPy could support numerical computation, signal integration, and statistical analysis of internal cognitive-state changes.

Scikit-learn could provide baseline anomaly-detection and classification techniques for comparison with the proposed monitoring approach.

NetworkX could support graph-based representations of relationships between goals, expectations, memory, and reasoning states.

These technologies would serve as implementation tools rather than defining the conceptual architecture itself.

7.3. Technical Limitations of Python

A major limitation is the computational cost of continuously monitoring multiple internal signals while the primary AI performs its tasks.

Python's interpreter overhead and limitations in CPU-bound parallel execution may affect performance in time-sensitive environments. However, optimized numerical libraries and GPU-supported frameworks can reduce some of these limitations.

Another challenge is accessing meaningful internal states from existing AI models. Some models expose useful intermediate information, while others provide only limited access to their internal processing.

Furthermore, developing a reliable Cognitive Tension mechanism would require distinguishing genuine warning patterns from harmless uncertainty, novelty, or normal variation.

These challenges are not exclusively Python limitations; they also reflect unresolved research questions in the proposed framework.

7.4. Alternative Programming Languages and Computing Platforms

For computationally demanding implementations, languages such as **C++ or Rust** could support performance-critical monitoring components and low-latency processing.

GPU-based computing platforms could accelerate the processing of multiple internal signals and the execution of diagnostic models.

Distributed computing architectures may also be useful when PEIF is integrated into large-scale AI systems that require continuous monitoring across multiple components.

However, these alternatives would become relevant primarily during advanced implementation stages. Python remains a practical choice for initial research and prototyping.

7.5. Future Technologies and Research Requirements

More advanced implementations of PEIF may benefit from developments in continual learning, efficient uncertainty estimation, interpretable internal representations, and adaptive resource allocation.

Neuromorphic computing, spiking neural networks, and other brain-inspired computing approaches could also be investigated as possible methods for efficient continuous monitoring. These technologies are potential research directions rather than mandatory components of PEIF.

Future research must determine how Cognitive Tension should be represented, which internal signals are most informative, and how the system can learn from error traces without introducing excessive false warnings.

Experimental studies should compare PEIF-enabled systems against conventional verification, uncertainty estimation, and anomaly-detection approaches using metrics such as early-warning lead time, detection rate, false-alarm rate, diagnostic success, and computational overhead.

Overall, PEIF appears conceptually suitable for initial prototype investigation using existing AI technologies. Nevertheless, its practical effectiveness, scalability, and advantages over current approaches remain unverified and require systematic experimental evaluation.

8. Part F – Ethical, Social and Research Implications

8.1. Potential Benefits

The proposed **Preconscious Error Intelligence Framework (PEIF)** could improve the reliability and safety of future AI systems by allowing them to detect signs of possible internal failure **before the exact error is identified**. Instead of waiting for an incorrect output or external feedback, the system could recognize unusual patterns within its own cognitive processing and direct additional attention toward them.

This capability could be particularly valuable in **high-risk applications** such as healthcare, finance, autonomous systems, and industrial decision-making, where an unnoticed error could have serious consequences. An early internal warning could encourage the AI to re-evaluate its reasoning, request additional information, or delay a decision when necessary.

PEIF could also support **continuous learning and adaptation**. By remembering internal patterns that occurred before previous errors, the system could become more sensitive to similar situations in the future. Over time, this may allow AI systems to improve not only from the errors they make, but also from the internal warning signs that appear before those errors.

Therefore, the main potential benefits of PEIF include **improved reliability, earlier error awareness, safer decision-making, better adaptability, and stronger self-monitoring capabilities** in future AI systems.

8.2. Ethical Considerations

The introduction of **Preconscious Error Intelligence Framework (PEIF)** raises several ethical considerations, particularly when it is used in high-impact decision-making. One major issue is **accountability**. If an AI generates an internal warning that something may be wrong but still produces an incorrect decision, it must be clear whether responsibility belongs to the AI developer, system operator, organization, or human decision-maker.

Transparency and explainability are also important. A cognitive-tension warning alone should not be treated as evidence that an error definitely exists. The system should clearly communicate the level of uncertainty and, where possible, explain what internal signals triggered further investigation.

Human oversight should remain important, especially in areas such as healthcare, finance, and autonomous systems. PEIF should support human decision-making rather than allowing unexplained internal warnings to automatically control critical decisions.

Finally, monitoring and storing internal error patterns may introduce **privacy and data-governance concerns**, particularly if these cognitive states are influenced by sensitive user information.

Appropriate controls would therefore be required to ensure that PEIF operates responsibly, transparently, and with clear human accountability.

8.3. Societal Impact

The **Preconscious Error Intelligence Framework (PEIF)** could have a positive societal impact by making future AI systems more reliable in areas where incorrect decisions can directly affect people. In fields such as **healthcare, finance, transportation, education, and public services**, the ability to recognize possible internal problems before making or finalizing a decision could reduce harmful AI errors.

PEIF could also improve **human-AI collaboration**. Instead of presenting every output with the same level of certainty, an AI system could indicate when its own cognitive process has produced a significant internal warning. This could help users recognize when additional human review or information is required.

However, the introduction of this capability may also influence how society **trusts AI systems**. People may place greater trust in systems that appear capable of monitoring themselves, which makes it important to communicate that an internal warning mechanism does not make an AI error-free or equivalent to human consciousness.

Therefore, the societal value of PEIF would depend not only on improving AI reliability, but also on ensuring that its capabilities and limitations are clearly understood by the people and organizations using it.

8.4. Possible Risks

Although the **Preconscious Error Intelligence Framework (PEIF)** could improve AI reliability, it may also introduce new risks. One major risk is **false warnings**, where the system develops high cognitive tension even though its original reasoning is correct. Frequent false warnings could cause unnecessary re-evaluation, slower decisions, and increased computational cost.

The opposite problem is **missed errors**. PEIF may fail to generate a warning when an actual error is developing, particularly when the system encounters unfamiliar situations or error patterns that it has not previously learned. Therefore, the framework should never be assumed to detect every possible error.

Another risk is **over-reliance on self-monitoring**. Users may trust an AI decision simply because the system reports no internal warning. However, the absence of cognitive tension does not guarantee that the decision is correct. Similarly, a warning should not automatically be interpreted as proof of an error.

Continuous monitoring of internal cognitive states may also require additional **memory, processing power, and computational resources**. In high-risk applications, incorrect interpretation or manipulation of the monitoring mechanism could also affect important decisions. Therefore, PEIF would require careful evaluation, security mechanisms, and appropriate human oversight before being used in critical real-world systems.

8.5. Future Research Opportunities

The **Preconscious Error Intelligence Framework (PEIF)** introduces several opportunities for future research. One important direction is investigating how **Cognitive Tension** could be computationally represented by combining weak internal signals such as prediction instability, uncertainty, memory conflicts, expectation mismatches, and other changes in internal cognitive states.

Future studies could also investigate how an AI system could learn **pre-error patterns from experience**. This includes determining whether internal states that occurred before previous failures can be stored and used to recognize similar situations before future errors become explicit.

Another important research direction is developing methods to evaluate whether PEIF genuinely provides **pre-diagnostic error detection**, rather than simply functioning as another anomaly detector or uncertainty estimator. Experiments could compare systems with and without PEIF by measuring early error detection, false warnings, missed errors, correction success, and computational cost.

Finally, future research could explore whether emerging computing approaches could support continuous internal monitoring more efficiently. Interdisciplinary research involving **Artificial Intelligence, Cognitive Science, Psychology, and Neuroscience** would also be valuable for understanding how biological error-monitoring mechanisms could inspire further development of PEIF.

9. Conclusion

This research investigated **Unconscious Error Detection**, a human cognitive capability that allows individuals to recognize that something may be wrong before consciously identifying the exact mistake. By examining psychological and neuroscientific perspectives alongside the limitations of existing Artificial Intelligence, the study explored how this capability could inspire a new direction in AI research.

The proposed **Preconscious Error Intelligence Framework (PEIF)** introduces a conceptual architecture that integrates continuous internal cognitive-state monitoring, expectation-based coherence evaluation, and a pre-diagnostic warning mechanism known as **Cognitive Tension**. This mechanism is designed to trigger adaptive attention and diagnostic reasoning when potential cognitive instability is detected, while **Error Trace Memory** supports experience-based learning and future adaptation.

Unlike conventional approaches focused primarily on task performance and explicit error verification, PEIF proposes an integrated process for sensing possible internal problems before their exact causes are identified. This approach could potentially improve AI reliability, adaptability, and decision-making in complex environments.

However, PEIF remains a **conceptual research proposal**. Its computational mechanisms, practical feasibility, and effectiveness require further investigation and experimental validation. Future research should focus on developing prototypes, evaluating early-warning capabilities, and comparing PEIF with existing error-monitoring approaches.

Overall, PEIF presents a proposed direction toward more self-monitoring and adaptive Artificial Intelligence inspired by human cognitive processes.

IT3071 – Machine Learning and Optimization Method (MLOM)

PRECONSCIOUS ERROR INTELLIGENCE FRAMEWORK (PEIF)

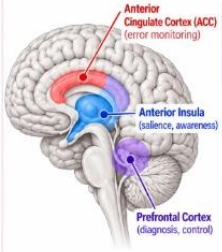
Next Intelligence – Designing the Next Generation of Artificial Intelligence Inspired by the Human Mind

Towards AI That Detects Something Is Wrong Before Knowing Why

1. SELECTED HUMAN COGNITIVE CAPABILITY

Unconscious Error Detection

"An agent senses that something is wrong before identifying the exact error."



Human brain process (simplified):

- Action / Prediction
- Mismatch detected (~50–100 ms) – ERN
- "Something is wrong" (pre-conscious signal)
- Conscious awareness (~200–400 ms) – Pe
- Identify error & correct

The human brain continuously monitors actions and outcomes, generating an early warning signal (ERN) before we consciously know the exact mistake.

2. LIMITATIONS OF CURRENT AI SYSTEMS

Current AI has related mechanisms, but key limitations prevent true preconscious error detection.

Supervised Learning (loss function)
- Detects error after output is produced
- Requires explicit ground truth

Reinforcement Learning
- Uses prediction error for learning
- No internal awareness signal

LLMs and Self-reflection
- Requires explicit prompting
- Not continuous internal monitoring

Anomaly Detection
- Detects abnormal inputs/outputs
- Not self-monitoring of reasoning process

Key Gap
No existing AI system genuinely possesses an internal, pre-diagnostic "something is wrong" signal during its own reasoning process.

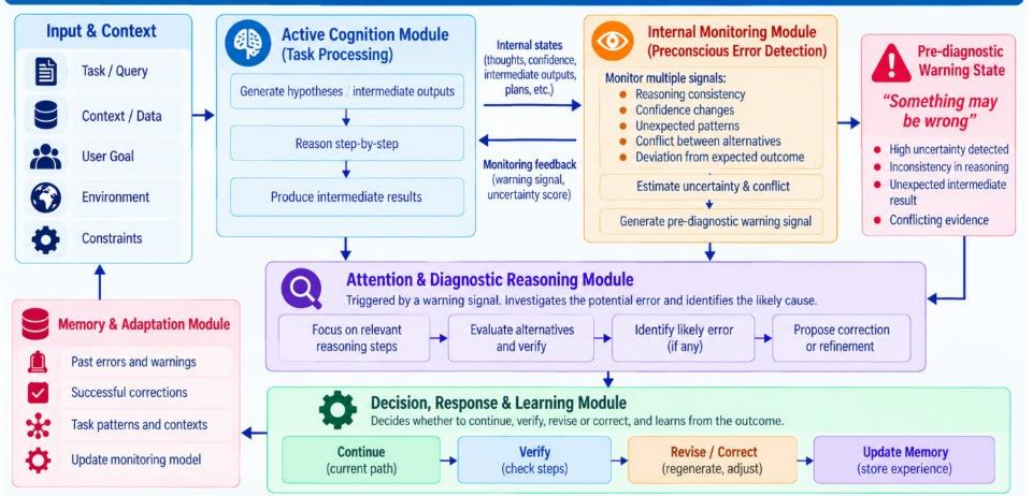
3. OUR PROPOSED NEXT GENERATION AI FRAMEWORK

Preconscious Error Intelligence Framework (PEIF)
A brain-inspired framework that continuously monitors its own cognitive process, generates an early warning state when potential errors emerge, and uses this signal to guide investigation and correction.

Core Principles

1. **Continuous Internal Monitoring** – Monitors reasoning, confidence, and consistency signals while performing a task.
2. **Pre-diagnostic Warning** – Generates an early internal state when something may be wrong, before exact diagnosis.
3. **Triggered Investigation** – Directs attention to relevant reasoning steps to identify the likely cause.
4. **Adaptive Learning** – Learns from past warnings and outcomes to improve future monitoring and decision-making.

4. CONCEPTUAL ARCHITECTURE OF PEIF



5. FEASIBILITY OF THE PROPOSED APPROACH

Conceptual feasibility
PEIF could be explored through a research prototype: its pre-diagnostic warning mechanism remains a proposal requiring validation.

1. **Implementation Possibility**
   - A conceptual prototype could be developed in Python.
   - Test monitoring signals on controlled reasoning tasks.
2. **Potential Tools**
   - PyTorch or JAX for modelling and experiments.
   - NumPy for signal analysis and evaluation.
3. **Technical Challenges**
   - Define measurable warning signals.
   - Reduce false alarms and missed errors.
   - Evaluate without always knowing the correct answer.
4. **Future Technologies**
   - Neuromorphic computing and spiking neural networks may be explored.
   - Specialised hardware is a future possibility, not a current requirement.

Feasibility Outlook
Start with a small simulation, measure warning quality, and compare it with established uncertainty and self-checking methods.

6. FUTURE RESEARCH DIRECTIONS

Several promising research directions can further enhance and validate the PEIF framework.

Neuroscience-inspired Models
- Integrate cognitive and neural mechanisms (e.g., ERN, ACC) into AI models.
- Explore biologically plausible architectures such as spiking neural networks.

Real-world Evaluation
- Test the framework in real-world domains (e.g., healthcare, autonomous systems).
- Measure reliability, robustness, and generalizability across diverse tasks.

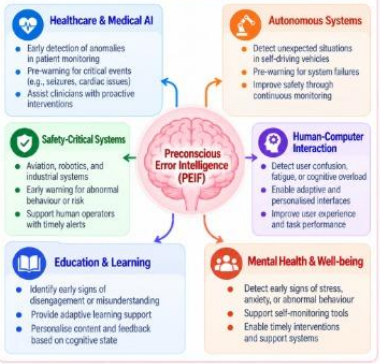
Adaptive and Continual Learning
- Develop adaptive mechanisms that evolve with experience.
- Improve long-term monitoring and reduce false alarms using meta-learning and continual learning techniques.

Human-AI Collaboration
- Investigate how preconscious warnings can be communicated effectively to users.
- Study trust, transparency, and user acceptance in human-AI teaming.

Long-term Vision
Move towards more trustworthy, self-monitoring and human-aligned AI systems that can proactively detect and mitigate errors in complex, real-world environments.

7. POTENTIAL APPLICATIONS

PEIF can be applied across multiple domains where early error detection and adaptive response are valuable.



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